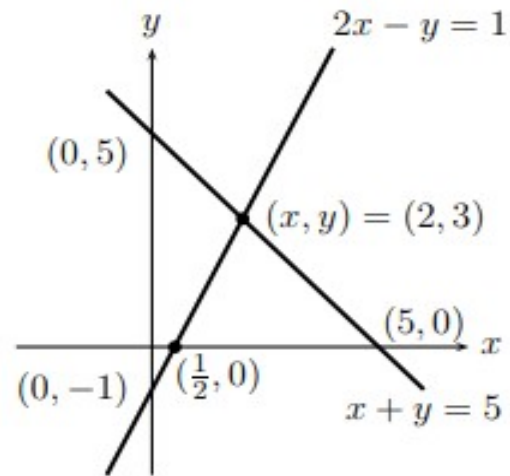
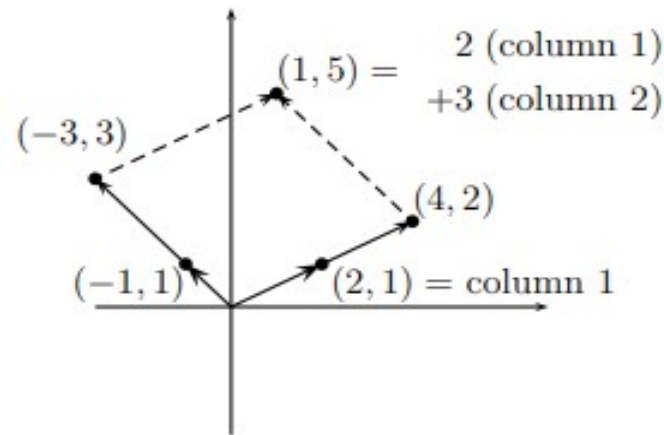


The geometry of linear equations

Linear System:

$$\begin{aligned} 2x - y &= 1 \\ x + y &= 5. \end{aligned}$$


(a) Lines meet at $x = 2, y = 3$



(b) Columns combine with 2 and 3

Figure : Row picture (two lines) and column picture (combine columns).

Column form

$$x \begin{bmatrix} 2 \\ 1 \end{bmatrix} + y \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

To find the combination of the column vectors on the left side that produces the vector on the right side

The geometry of linear equations

Linear System:

$$\begin{aligned}2u + v + w &= 5 \\4u - 6v &= -2 \\-2u + 7v + 2w &= 9.\end{aligned}$$

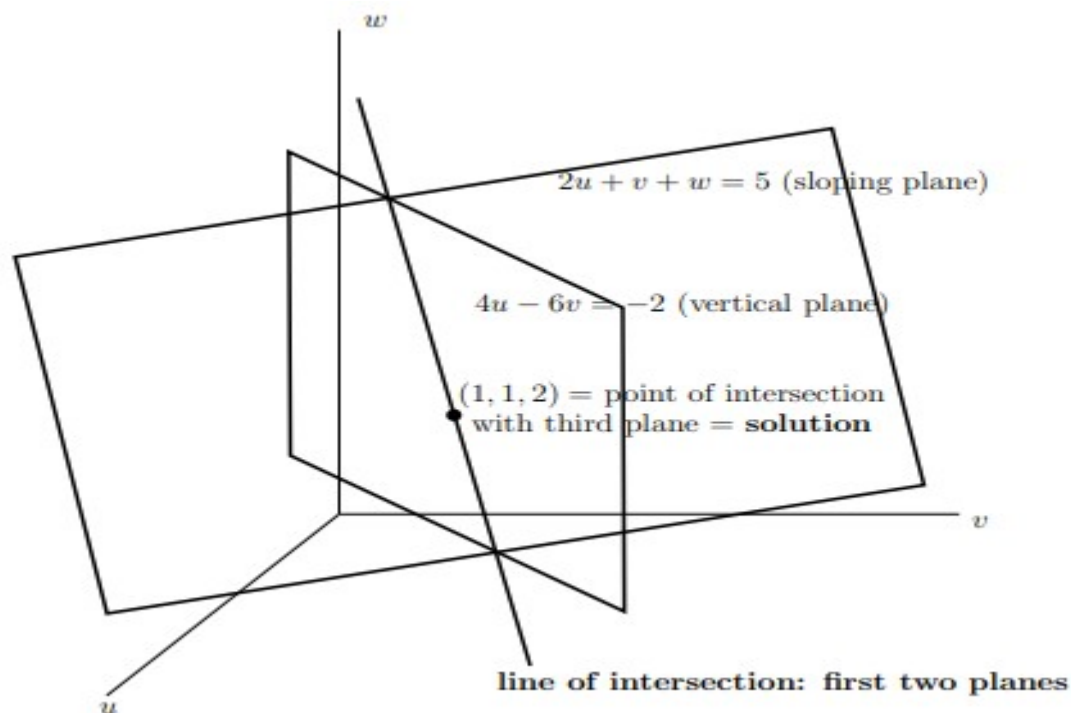


Figure : The row picture: three intersecting planes from three linear equations.

The first equation produces an $(n-1)$ -dimensional plane in n dimensions, The second plane intersects it (we hope) in a smaller set of dimension $n-2$. Assuming all goes well, every new plane (every new equation) reduces the dimension by one. At the end, when all n planes are accounted for, the intersection has dimension zero. It is a point, it lies on all the planes, and its coordinates satisfy all n equations. It is the solution!

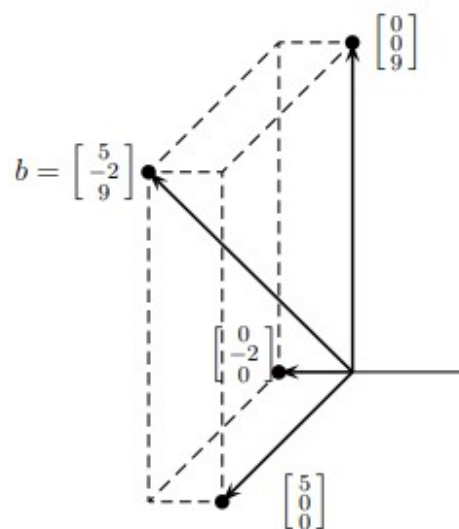
The geometry of linear equations

Linear System:

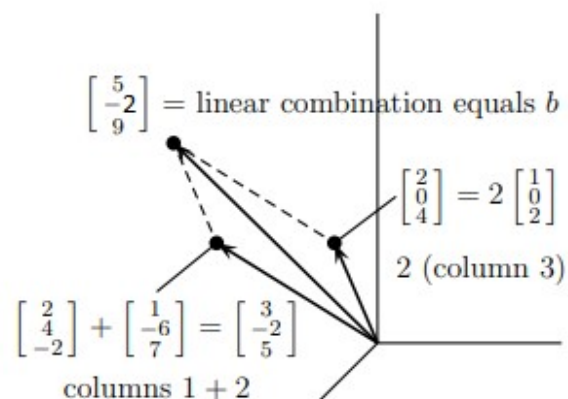
$$\begin{aligned} 2u + v + w &= 5 \\ 4u - 6v &= -2 \\ -2u + 7v + 2w &= 9. \end{aligned}$$

Column form

$$u \begin{bmatrix} 2 \\ 4 \\ -2 \end{bmatrix} + v \begin{bmatrix} 1 \\ -6 \\ 7 \end{bmatrix} + w \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix}$$



(a) Add vectors along axes



(b) Add columns $1 + 2 + (2 \times 3)$

Figure : The column picture: linear combination of columns equals b .

Row picture: Intersection of planes

Column picture: Combination of columns

The geometry of linear equations

With n equations in n unknowns, there are n planes in the row picture. There are n vectors in the column picture, plus a vector b on the right side. The equations ask for a linear combination of the n columns that equals b . For certain equations that will be impossible. We look at the geometry exactly when it breaks down, in the singular case.

The Singular Case

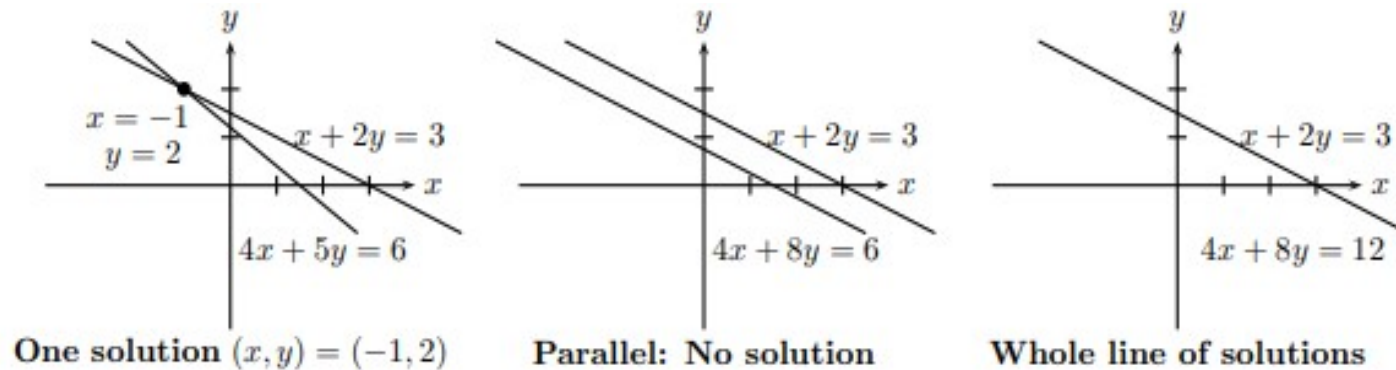


Figure : The example has one solution. Singular cases have none or too many.

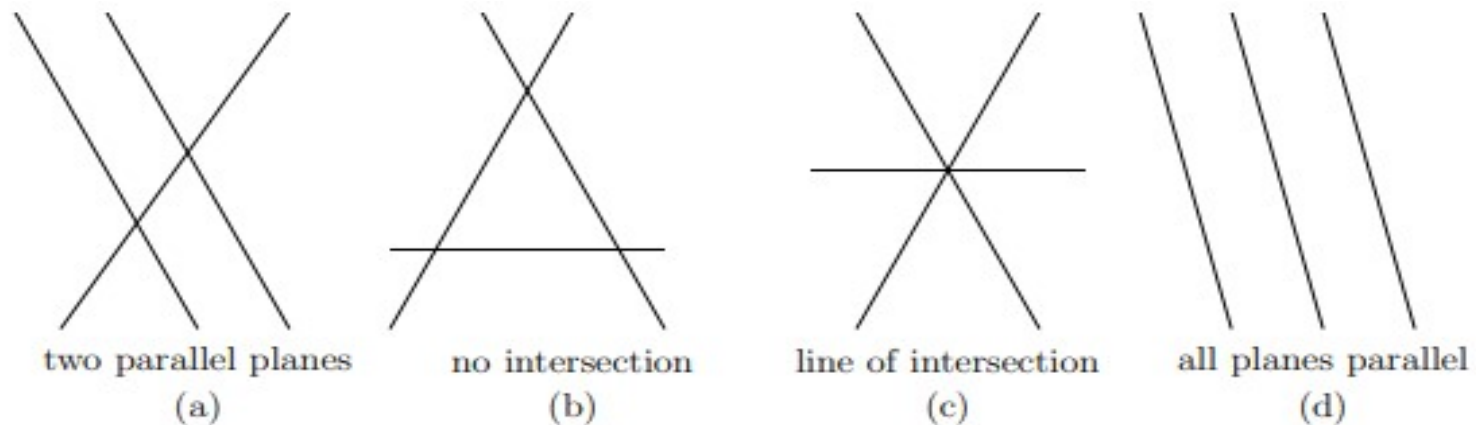


Figure : Singular cases: no solution for (a), (b), or (d), an infinity of solutions for (c).

The geometry of linear equations

The Singular Case

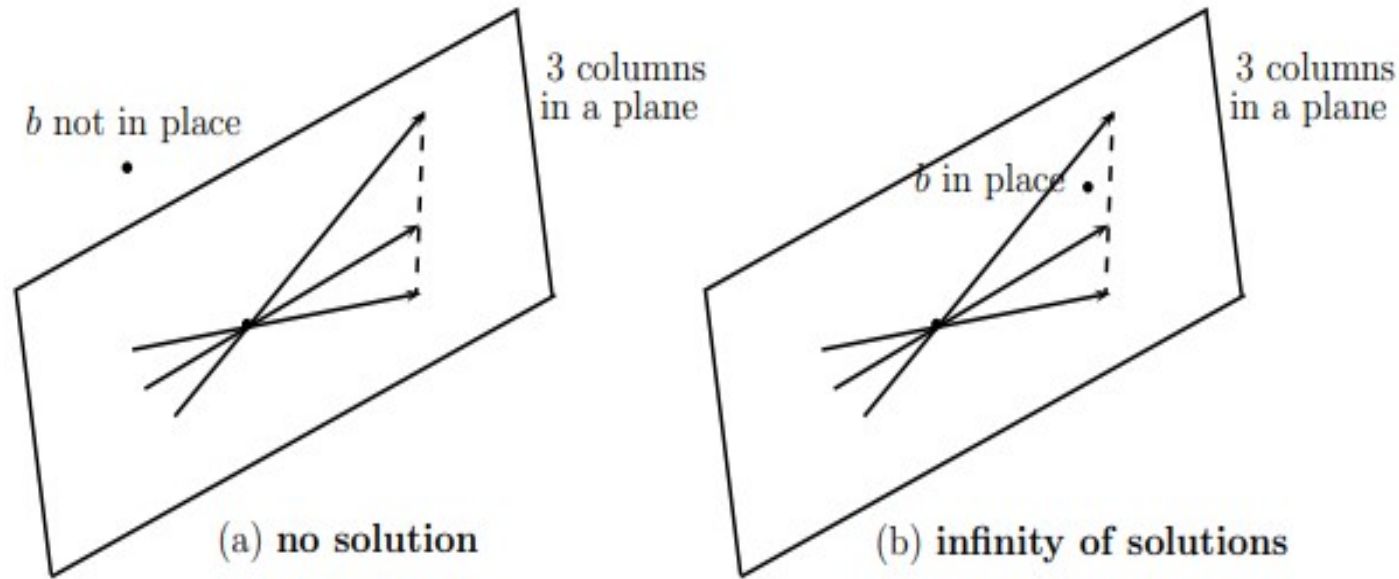


Figure : Singular cases: b outside or inside the plane with all three columns.

If the vector b is not in that plane, no solution is possible. That is by far the most likely event; a singular system generally has no solution. But there is a chance that b does lie in the plane of the columns. In that case there are too many solutions; the three columns can be combined in infinitely many ways to produce b .

Gaussian Elimination

$$\begin{array}{lcl} \text{Original system} & 2u & + v + w = 5 \\ & 4u & - 6v = -2 \\ & -2u & + 7v + 2w = 9 \end{array}$$

Forward elimination: The method starts by subtracting multiples of the first equation from the other equations.

$$\left[\begin{array}{cccc} 2 & 1 & 1 & 5 \\ 4 & -6 & 0 & -2 \\ -2 & 7 & 2 & 9 \end{array} \right] \longrightarrow \left[\begin{array}{cccc} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 8 & 3 & 14 \end{array} \right] \longrightarrow \left[\begin{array}{cccc} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

Back-substitution:

$$\begin{array}{rcl} 2u & + & v + w = 5 \\ & - & 8v - 2w = -12 \\ & & 1w = 2 \end{array} \qquad w=2, v=1, u=1$$

By definition, pivots cannot be zero. With a full set of n pivots, there is only one solution. The system is non singular, and it is solved by forward elimination and back-substitution. But if a zero appears in a pivot position, elimination has to stop—either temporarily or permanently. The system might or might not be singular. Notice that during elimination process a zero can appear in a pivot position, even if the original coefficient in that place was not zero. In many cases this problem can be cured, and elimination can proceed. Such a system still counts as non singular; it is only the algorithm that needs repair. In other cases a breakdown is unavoidable. Those incurable systems are singular, they have no solution or else infinitely many, and a full set of pivots cannot be found.

Gaussian Elimination

Example 1. Nonsingular (cured by exchanging equations 2 and 3)

$$\begin{array}{rcl} u + v + w = _ & u + v + w = _ & u + v + w = _ \\ 2u + 2v + 5w = _ \rightarrow & 3w = _ \rightarrow & 2v + 4w = _ \\ 4u + 6v + 8w = _ & 2v + 4w = _ & 3w = _ \end{array}$$

The system is now triangular, and back-substitution will solve it.

Example 2. Singular (incurable)

$$\begin{array}{rcl} u + v + w = _ & u + v + w = _ \\ 2u + 2v + 5w = _ \rightarrow & 3w = _ \\ 4u + 4v + 8w = _ & 4w = _ \end{array}$$

When Gaussian Elimination process breakdown?

Answer: If zero is in pivot position and not possible to exchange then it fails

Matrix multiplication

Matrix notation:

	$2u + v + w = 5$	Nine coefficients
Linear System:	$4u - 6v = -2$	Three unknowns
	$-2u + 7v + 2w = 9$	Three right-hand sides

This linear system can be describe using matrix notation as follows: $Ax=b$, where A is 3x3 coefficient matrix, x is unknown column vector and b is right-hand side column vector.

Matrix form $Ax = b$

$$\begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix}$$

Matrix multiplication

Method 1:

$$\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 1 \times 1 + 2 \times 4 & 1 \times 2 + 2 \times 5 & 1 \times 3 + 2 \times 6 \\ 0 \times 1 + 3 \times 4 & 0 \times 2 + 3 \times 5 & 0 \times 3 + 3 \times 6 \end{bmatrix} = \begin{bmatrix} 9 & 12 & 15 \\ 12 & 15 & 18 \end{bmatrix}$$

Method 2: Row method: Linear combination of rows

$$\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 1[1 \ 2 \ 3] + 2[4 \ 5 \ 6] \\ 0[1 \ 2 \ 3] + 3[4 \ 5 \ 6] \end{bmatrix} = \begin{bmatrix} [1 \ 2 \ 3] + [8 \ 10 \ 12] \\ [0 \ 0 \ 0] + [12 \ 15 \ 18] \end{bmatrix} = \begin{bmatrix} 9 & 12 & 15 \\ 12 & 15 & 18 \end{bmatrix}$$

Matrix multiplication

Method 3: Column method: Linear combination of columns

$$\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \left[1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 4 \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 5 \begin{bmatrix} 2 \\ 3 \end{bmatrix} \quad 3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 6 \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right] = \begin{bmatrix} 9 & 12 & 15 \\ 12 & 15 & 18 \end{bmatrix}$$

Method 4: Sum of Columns x rows

$$\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \left[\begin{bmatrix} 1 \\ 0 \end{bmatrix} [1 \ 2 \ 3] + \begin{bmatrix} 2 \\ 3 \end{bmatrix} [4 \ 5 \ 6] \right] = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 8 & 10 & 12 \\ 12 & 15 & 18 \end{bmatrix} = \begin{bmatrix} 9 & 12 & 15 \\ 12 & 15 & 18 \end{bmatrix}$$

Example: Find the matrix that exchange row 1 and row 2 of $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Example: Find the matrix that exchange column 1 and column 2 of $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

Example: Find the permutation matrices that exchange rows of a 3x3 matrix

Triangular factors

Original system

$$\begin{array}{rrcr} 2u & + & v & + & w & = & 5 \\ 4u & - & 6v & & & = & -2 \\ -2u & + & 7v & + & 2w & = & 9 \end{array} \quad \longrightarrow \quad Ax = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ 9 \end{bmatrix} = b$$

Step 1. Subtract 2 times the first equation from the second;

Step 2. Subtract -1 times the first equation from the third;

Step 3. Subtract -1 times the second equation from the third.

$$Ux = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 5 \\ -12 \\ 2 \end{bmatrix} = c$$

Elementary matrices:

For Step 1, $E = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

For Step 2, $F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

For Step 3, $G = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$

The result of all three steps is $GFE = U$. We could multiply GFE together to find the single matrix that takes A to U (and also takes b to c). It is lower triangular:

From A to U $GFE = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$

Triangular factors

How would we get from U back to A? How can we undo the steps of Gaussian elimination?

To undo step 1 is not hard. Instead of subtracting, we add twice the first row to the second. The result of doing both the subtraction and the addition is to bring back the identity matrix:

$$\begin{array}{l} \text{Inverse of} \\ \text{subtraction} \\ \text{is addition} \end{array} \quad \begin{array}{c} E^{-1} \\ \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array} \quad \begin{array}{c} E \\ \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array} = \begin{array}{c} I \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array}$$

Since step 3 was last in going from A to U, its matrix G must be the first to be inverted in the reverse direction. Inverses come in the opposite order!

$$\begin{array}{l} \text{From } U \text{ back to } A \\ E^{-1}F^{-1}G^{-1}U = A \\ LU = A. \end{array}$$

Recognize the matrix L that takes U back to A:

$$E^{-1}F^{-1}G^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{bmatrix} = L$$

Triangular factorization $A = LU$ with no exchanges of rows. L is lower triangular, with 1s on the diagonal. The multipliers (taken from elimination) are below the diagonal. U is the upper triangular matrix which appears after forward elimination, The diagonal entries of U are the pivots.

Triangular factors

Example 1.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix}$$

Example 2. (which needs a row exchange)

$$A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix} \quad \text{cannot be factored into } A = LU.$$

Example 3. (with all pivots and multipliers equal to 1)

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

Example 4. (when U is the identity and L is the same as A)

$$A = \begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix}.$$

Inverses

The inverse of an n by n matrix is another n by n matrix. The inverse of A is written A^{-1} . If you multiply by A and then multiply by A^{-1} , you are back where you started:

$$\text{If } b = Ax \text{ then } A^{-1}b = x.$$

The inverse of A is a matrix B such that $BA = I$ and $AB = I$. There is at most one such B , and it is denoted by A^{-1} : $A^{-1}A = I$ and $AA^{-1} = I$.

- The inverse exists if and only if elimination produces n pivots.
- The matrix A cannot have two different inverses.

Suppose A has two inverse B and C . $B(AC) = (BA)C \Rightarrow BI = IC$ which is $B = C$.

- Suppose there is a nonzero vector x such that $Ax = 0$. Then A cannot have an inverse.

$$Ax = 0 \Rightarrow A^{-1}Ax = A^{-1}0 \Rightarrow x = 0 \text{ which is not possible.}$$

- A diagonal matrix has an inverse provided no diagonal entries are zero:

$$\text{If } A = \begin{bmatrix} d_1 & & \\ & \ddots & \\ & & d_n \end{bmatrix} \text{ then } A^{-1} = \begin{bmatrix} 1/d_1 & & \\ & \ddots & \\ & & 1/d_n \end{bmatrix} \text{ and } AA^{-1} = I.$$

- A product AB of invertible matrices is inverted by $B^{-1}A^{-1}$:

Proof. To show that $B^{-1}A^{-1}$ is the inverse of AB , we multiply them and use the associative law to remove parentheses. Notice how B sits next to B^{-1} :

$$(AB)(B^{-1}A^{-1}) = ABB^{-1}A^{-1} = AIA^{-1} = AA^{-1} = I$$

$$(B^{-1}A^{-1})(AB) = B^{-1}A^{-1}AB = B^{-1}IB = B^{-1}B = I.$$

A similar rule holds with three or more matrices:

$$\text{Inverse of } ABC \quad (ABC)^{-1} = C^{-1}B^{-1}A^{-1}.$$

The Calculation of A^{-1} : The Gauss-Jordan Method

$$\begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix} \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \overset{\mathbf{A}^{-1}}{=} \begin{bmatrix} e_1 & e_2 & e_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$Ax_1 = e_1, Ax_2 = e_2 \text{ and } Ax_3 = e_3$$

Thus we have three systems of equations (or n systems). They all have the same coefficient matrix A . The right-hand sides e_1, e_2, e_3 are different, but elimination is possible *on all systems simultaneously*. This is the **Gauss-Jordan method**. Instead of stopping at U and switching to back-substitution, it continues by subtracting multiples of a row *from the rows above*. This produces zeros above the diagonal as well as below. When it reaches the identity matrix we have found A^{-1} .

$$\begin{aligned} [A : I] &= [GFEA : GFEI] = [U : L^{-1}] = [U^{-1}U : U^{-1}L^{-1}] \\ &= [I : (LU)^{-1}] = [I : A^{-1}] \end{aligned}$$

The Calculation of A^{-1} : The Gauss-Jordan Method

Example 1. Using the Gauss-Jordan Method to Find A^{-1}

$$\begin{bmatrix} A & e_1 & e_2 & e_3 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 & 1 & 0 & 0 \\ 4 & -6 & 0 & 0 & 1 & 0 \\ -2 & 7 & 2 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{pivot} = 2 \rightarrow \begin{bmatrix} 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & -8 & -2 & -2 & 1 & 0 \\ 0 & 8 & 3 & 1 & 0 & 1 \end{bmatrix}$$

$$\text{pivot} = -8 \rightarrow \begin{bmatrix} 2 & 1 & 1 & 1 & 0 & 0 \\ 0 & -8 & -2 & -2 & 1 & 0 \\ 0 & 0 & 1 & -1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} U & L^{-1} \end{bmatrix}$$

$$\text{Second half} \quad \begin{bmatrix} U & L^{-1} \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & 0 & 2 & -1 & -1 \\ 0 & -8 & 0 & -4 & 3 & 2 \\ 0 & 0 & 1 & -1 & 1 & 1 \end{bmatrix}$$

$$\text{zeros above pivots} \rightarrow \begin{bmatrix} 2 & 0 & 0 & \frac{12}{8} & -\frac{5}{8} & -\frac{6}{8} \\ 0 & -8 & 0 & -4 & 3 & 2 \\ 0 & 0 & 1 & -1 & 1 & 1 \end{bmatrix}$$

$$\text{divide by pivots} \rightarrow \begin{bmatrix} 1 & 0 & 0 & \frac{12}{16} & -\frac{5}{16} & -\frac{6}{16} \\ 0 & 1 & 0 & \frac{4}{8} & -\frac{3}{8} & -\frac{2}{8} \\ 0 & 0 & 1 & -1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} I & A^{-1} \end{bmatrix}$$

Transpose and Symmetric Matrices

Transpose If $A = \begin{bmatrix} 2 & 1 & 4 \\ 0 & 0 & 3 \end{bmatrix}$ then $A^T = \begin{bmatrix} 2 & 0 \\ 1 & 0 \\ 4 & 3 \end{bmatrix}$

(i) The transpose of AB is $(AB)^T = B^T A^T$,

Start from $AB = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 3 & 3 \\ 2 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 3 & 3 \\ 5 & 5 & 5 \end{bmatrix}$

Transpose to $B^T A^T = \begin{bmatrix} 3 & 2 \\ 3 & 2 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 3 & 5 \\ 3 & 5 \end{bmatrix}$.

(ii) The transpose of A^{-1} is $(A^{-1})^T = (A^T)^{-1}$.

We know that $A^{-1}A = AA^{-1} = I$, Taking transpose of these terms we will get:

$$(A^{-1}A)^T = (AA^{-1})^T = I \quad \longrightarrow \quad A^T (A^{-1})^T = (A^{-1})^T A^T = I. \text{ So, inverse of } A^T \text{ is } (A^{-1})^T \text{ i.e., } (A^T)^{-1} = (A^{-1})^T$$

A symmetric matrix is a matrix that equals its own transpose i.e., $A^T = A$.

Symmetric matrices $A = \begin{bmatrix} 1 & 2 \\ 2 & 8 \end{bmatrix}$ and $D = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$

(i) *For any matrix R , the product $R^T R$ and RR^T is a square symmetric matrix:*